

PPPG-DialoGPT: A Prompt-based and Personality-aware Framework For Conversational Recommendation Systems

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Abstract—Conversational Recommendation Systems (CRSs) are multi-turn dialogue softwares that support recommendation goals. Having a CRS that generates personalized recommendations for its users is one of the challenging topics in the recommendation research field. This is due to the diversity of preferences that a user might have within a recommendation session. Recent CRS frameworks are still not really able to effectively provide items of interest to their users, as they do not consider the individuals' personality traits during the recommendation process. These traits are the enduring characteristics and behaviors that comprise a person's unique adjustment and distinguish him from other persons. Thus, considering them during the recommendation process could significantly boost the model performance. In this paper, we propose a personality-aware and prompt-based CRS framework named PPPG-DialoGPT "A Personality and Preference-aware Prompt-based Goal-oriented DialoGPT model". Our proposed approach aims to benefit from the significant impact that personality traits and prompt-based learning frameworks have on improving the performance of different Natural Language Processing (NLP) and one-shot recommendation tasks. To the best of our knowledge, this paper is the first to employ both individual personality traits and prompts templates during a dialogue-based recommendation session in the field of conversational recommendation system. Experiments and results show that the use of the users' personality traits leads to an improvement in the performance of both recommendation and response generation tasks on the movie-based TG-Redial dataset.

Index Terms—Conversational Recommendation System, Prompt-based learning, Personality traits, Language models.

I. INTRODUCTION

Traditional Recommendation Systems (RSs) often rely on a *one-shot* interaction paradigm to predict the item of interest for a specific user. While these systems can be effective in some scenarios, they often fail to provide personalized recommendations. This is because they consider the user's past observed behaviors as the ground truth to build and understand their current preferences which is generally not the case [1]. To overcome these drawbacks, the CRS paradigm was proposed to the literature [2]. Unlike traditional one-shot systems, CRSs employ both previous and current user preferences during the recommendation process [2]. With the recent advancement in *Natural Language Processing*

(NLP) and deep learning technologies, the interest in CRS during the past few years has increased significantly. Different promising approaches have been proposed to the literature to implement *end-to-end* CRS. A recent line of work focuses on using pre-trained transformer models such as *GPT-2* [3] and *BERT* [4] by fine-tuning them to produce specific and personalized recommended items [5]. Despite the promising results provided by these models on the *topic-oriented* conversational recommendation task [6], [7], these models are originally trained on large unstructured open-domain textual datasets (books scripts, movies scripts, open-domain internet information, etc...). Therefore, they do not encode the *dialogue turns structure* within their parameters. Moreover, training or fine-tuning these types of models is shown to be computationally expensive. To efficiently train and fine-tune such huge models, a new learning paradigm called *Prompt-based learning* [8] has been proposed in the literature.

Prompt-based learning is a *self-supervised* learning paradigm proposed to solve the efficiency issues within the pre-trained *Language Models* (LM). In this paradigm, a learned or manually crafted *template* is added to create a new representation for the original input text to prompt the knowledge of a pre-trained LM. Instead of adapting and designing different model architectures and different objectives to fit the existing data, we adapt the data to fit an existing model. Recent research shows significant improvements using the prompt-based learning paradigm for different tasks such as the recommendation problem [9], [10]. However, previous research is done only on the one-shot recommendation frameworks.

Moreover, personality traits refer to the difference among individuals' characteristic patterns, such as how they think, feel, and behave. In the past few years, different studies proved that understanding people's personality traits and knowing what they are good at can be very important in a wide variety of situations such as improving the recommendation frameworks [1], [11]–[13]. The *Five-Factor* personality (Big5) system [14] measures the human personality along five distinct and independent dimensions: *Extroversion* (EXT), *Neuroticism* (NEU), *Agreeableness* (AGR), *Conscientiousness*

(CSN), and *Openness* (OPN). Extroversion refers to sociability and high amounts of emotional expressiveness. Neuroticism refers to moodiness and emotional instability. Agreeableness refers to trust, kindness, and other related behaviors. Conscientiousness refers to high levels of thoughtfulness and goal-directed behaviors. Openness refers to imagination and insight [15].

The use of these traits could improve the performance of CRSs [1] as they are considered as additional multi-contexts information. For example, introverted people usually talk less while extroverted people tend to talk more, so incorporating personality traits could help better model utterance turns. Raza *et al.* [16] underscores the significance of incorporating multi-context information such as time, user preferences, and news topics simultaneously in recommender systems, enabling the recommender system to provide a comprehensive and personalized user experience by considering multiple dimensions of relevance. To the best of our knowledge, there is currently no research that incorporates an individual's *Big5* personality traits factors as multi-contexts information into the conversational recommendation task as most of the existing *personality-aware* recommendation approaches are one-shot systems. As Golbeck *et al.* demonstrated a strong link between the individuals' personality traits and personalized recommended items [17], we believe that the absence of these traits could potentially contribute to the diminished effectiveness of CRS approaches in constructing personalized dialogue.

To this end, the novelty of our research resides in:

- Introducing a novel framework named **PPPG-DialoGPT** "A Personality and Preference-aware Prompt-based Goal-oriented DialoGPT model". This framework leverages the knowledge of the pre-trained model *DialoGPT* [18] through prompts to enhance the generation of natural conversation recommendations and responses. To elaborate, to enhance the personalization of the suggested items, the system considers the individual's personality traits as prompts to reshape the inputs of the *DialoGPT* model, thereby attaining a higher level of personalization and achieving a more accurate personality-aware recommender system, leading to improvements in both the efficiency and effectiveness of CRS.
- Investigating the impact of different control information types (users' personality traits, users' preferences, conversation topics, etc...) by manually designing different prompting templates and evaluating their performances on the task of recommendation using conversations.

The rest of this paper is organized as follows: section 2 describes the related work, section 3 presents the methodology and the prompting templates designing, section 4 discusses the experiments and results findings, and section 5 discusses the conclusions and future work.

II. BACKGROUND AND RELATED WORK

In this section, we first highlight the different conversational recommendation approaches that tend to be related to our research. Then, we introduce the *Big5* personality test and the recent recommendation systems related to this framework. Finally, we discuss the recently published prompt-based recommendation approaches that are related to our research.

A. Conversational Recommendation Systems

Conversational recommendation systems are types of multi-turn dialogue software that support recommendation goals for a specific user [2]. In recent years, the number of CRS published papers increased significantly [1], [2]. A new direction of research has been presented to the literature that combines the use of automatic learning and optimization approaches within a conversational recommendation goal [19], [20]. Rimond *et al.* [21] published a new CRS architecture that uses a combination of different neural network models (A Recurrent Neural Network (*RNN*) and an *AutoEncoder* network). Although the proposed architecture is highly effective, developing such approaches can be computationally expensive due to the need to train various modules on different objectives, which requires significant resources.

Kun Zhou *et al.* [6] used two pre-trained GPT-2 [3] models to generate topic-guided responses in the movie domain where answers were guided by the topics of the conversation. The authors considered two different scenarios for dialogue generation (a recommendation scenario and a non-recommendation one). Unlike Kun Zhou *et al.* [6] who used two separate scenarios, Hongru Wang *et al.* [7] proposed a new End-to-End network to achieve both topic prediction and response generation simultaneously. The authors proposed a learning refining approach for topic-aware dialogue generation by utilizing two GPT-2 models for dialogue generation and a BERT model for the topic prediction task.

Despite the improvement of both topic prediction and oriented response generation models' performance on the *TG-Redial* dataset [6], this approach is highly computationally expensive as it fine-tunes three large pre-trained LMs. Additionally, to control the generation process, the authors used only the topic information as control features, which could not be sufficient to generate personalized responses. Other features such as preferences and personalities could also be crucial factors in personalizing a conversation [1].

B. Personality traits in recommender systems

In the last few years, personality traits have been incorporated into various recommendation systems to improve their performance. Xiao *et al.* [23] proposed a new personality-aware system that recommends new followers based on text semantics and sentiment analysis. Their contribution resides in the combination of both the user's attributes and their *Big-Five* traits for the recommendation of followers. Recently a new work submitted by Souvikda *et al.* [36] tackles the controllable response generation task using individuals' *Big5* personality traits. Using both *Big5* personality traits and current utterance

intent/topic as control codes, the authors provide a better policy modeling for the response generator model, which results in a set of configurable parameters that can be modified to generate diverse personal responses. For the experimental setup, the authors used both *BART* [24] and *BlenderBot* [25] pre-trained models for the classification and response generation tasks respectively.

Despite the good performance provided by this approach, this model is only trained for dialogue purposes, but not for recommendation goals. Moreover, this architecture can be categorized within a *multi-objective* learning paradigm approach, where we have different loss functions and the model tries to perform effectively on all the tasks. The use of different objectives within the same model can be considered a drawback for these types of architectures, as it significantly increases the model's complexity by having more parameters for each objective loss.

To bypass the objective learning paradigm drawbacks, a new learning paradigm named *prompt-based learning* was proposed in the literature.

C. Prompt-based Learning Paradigm

Prompt-based learning has been successfully applied to many applications, such as text summarizing [26], text classification [27], text generation [3], and lately *item recommendation*. Lei *et al.* [10] proposed a prompt-based learning model named *Personalized Prompt Learning for Explainable Recommendation* (PEPLER) to generate explanation sentences for the recommendation systems using the user IDs and the item IDs as prompts. Despite the good recommendation performance provided by the PEPLER model, this approach focuses on the explanation problem rather than the recommendation task. During the experiments and results, the authors only reported the explanation task performance and they did not report the recommendation performance. Also, the prompts template was designed to fulfill the explanation purpose and not the recommendation one. Another recent work published by Sileo *et al.* [9] has also employed the prompt-based learning paradigm for the recommendation task. In their work, the authors used a manually designed prompt to formulate a zero-shot learning sequential movie recommendation task. They used the pre-trained GPT-2 model without fine-tuning and demonstrated better performance than matrix factorization approaches on the MAP@1 metric for a small number of users on the MovieLens 1M dataset. This demonstrates the potential of prompt-based learning in recommendation tasks.

Our work utilizes prompt-based learning within a recommendation framework, similar to Lei *et al.* [10]. However, our approach differs in that we focus on designing prompting templates for the recommendation task, rather than the explanation task. Furthermore, our work is similar to Hongru Wang *et al.* [7] in terms of utilizing topic information as controls for dialogue generation. However, our approach only involves a single language model with prompting templates,

as opposed to using three different language models. It is also important to highlight that the work proposed by Souvikda *et al.* [36] is very similar to our work in terms of using pre-trained language models and individuals' personalities for personalizing the response generation process. However, our work differs in terms of how we employed the generation control factors, as we focused on using the prompt-learning paradigm to efficiently produce more personalized responses, compared to the above Souvikda *et al.*'s work, which uses the objective learning paradigm.

III. THE PPPG-DIALOGPT MODEL

Throughout this section, we discuss our proposed PPPG-DialoGPT architecture. We start by stating the problem, then we highlight the design process of the prompting templates. Finally, we discuss the full architecture design of our proposed approach.

A. Problem Statement and Task Description

In an autoregressive language generation problem the probability of generating the next word at time t " $w_t \in V$ " depending on all of the previously generated words " $w_{1:t-1}$ " is defined as $p(w_t|w_{1:t-1})$, where V defines the language model vocabulary. In our framework, we consider the dialog-based recommendation process as a controllable language generation problem where the previous conditional generation probability is modified to include not only the previous words " $w_{1:t-1}$ " but also a set of controls named $C = \{S, R, T, P, B\}$ where S is the set of utterances, R is the set of the speaker's role, T is the set of topics mentioned in the conversation dialogue, $B = \{b_1, b_2, b_3, b_4, b_5\}$ is the user Big Five personality trait tokens and $P = \{p_1 \dots p_n\}$ represents the different words that define the user's preferences "profile" (see figure 1 for a toy example). Conditioning on the set of controls information C , the probability framework for generating the *next word* a time t can be designed as follows:

$$p(w_t|w_{1:t-1}, C_{1:t-1}) = \prod_{i=1}^{|x_t|} p(w_t^i|S_{1:t-1}, R_{1:t-1}, T_{1:t-1}, P, B, w_t^{1:i-1}) \quad (1)$$

where $t \in \mathbb{Z}^+$ is a positive number > 0 , where w_t^i is the current word to be generated at utterance t and position i , and $w_t^{1:i-1}$ are all the previously generated words within the utterance t , $S_{1:t-1}$ is the set of previous utterances within the conversation session, $R_{1:t-1}$ highlights the sequence of speakers role for the previous utterances, and $T_{1:t-1}$ is the sequence of preceding topics mentioned within the dialogue. Approximating this type of probability is very hard and computationally expensive as we are conditioning on too much information by multiplying different probabilities. However, using the prompt-based learning paradigm, we can simplify this problem definition to just modeling the probability $p(x)$. We reformulate the task of maximizing $p(x_t|S_{1:\{t-1\}}, R_{1:\{t-1\}}, T_{1:\{t-1\}}, P, B)$ to just maximizing the input probability $p(x)$, where $x_t = \|w_t^i\|$ is the sentence

at time t which contains the set of words at time t , x is a concatenation of all the $x_{1:t-1}$, $S_{1:t-1}$, $R_{1:t-1}$, $T_{1:t-1}$, P , and B information.

All types of information are considered as words. It is the learning model's responsibility to learn them in an autoregressive way. Therefore, our task setup for the probability of generating the target sentence at time t becomes as follows:

$$p(w_t|w_{1:t-1}, C_{1:t-1}) = \prod_{i=1}^N p(w_t^i|w_t^{1:i-1}) \quad (2)$$

To concatenate the control information within the same input, we used different prompting templates. The next section highlights the designing procedure of these templates.

B. Prompting Templates

Prompting is just about changing the input form using a specific template. Table I highlights our manually designed templates and their objectives.

TABLE I
THE PROMPTING TEMPLATES.

Prompt Name	Template	Objective
Promptless	[X][Z]	This prompt was designed as a baseline prompt.
T-prompt	[X] < topic > t ₁ , t ₂ , ..., t _n < /topic > < role > [Z]	We designed this prompt to evaluate the importance of adding topic information for the dialogue generation process compared to the Promptless template.
Pro-prompt	< profile > p ₁ , p ₂ , ..., p _n < /profile > [X] < topic > t ₁ , t ₂ , ..., t _n < /topic > < role > [Z]	We designed this prompt to evaluate the impact of adding the user's profile and it's preferences as controls for the dialogue generation process.
Pre-prompt	< big5 > per ₁ , per ₂ , ..., per ₅ < /big5 > [X] < topic > t ₁ , t ₂ , ..., t _n < /topic > < role > [Z]	We designed this prompt to evaluate the importance of adding the user's Big Five personality traits as controls for the dialogue generation process.
Pre-pro-prompt	< big5 > per ₁ , per ₂ , ..., per ₅ < /big5 > < profile > p ₁ , p ₂ , ..., p _n < /profile > [X] < topic > t ₁ , t ₂ , ..., t _n < /topic > < role > [Z]	We designed this prompt to investigate the impact of employing both the user's Big Five personality traits and it's preferences on our model performance

In our proposed templates, “[X]” is the original input that defines the previous conversation history. “[Z]” is the next utterance response given the original input “[X]”, “< topic >” and “< /topic >” are special tokens that define the start and the end of the topic information section. “t₁, t₂, ..., t_n” highlight the previous utterances topic information (the topic tokens), “< role >” = {< recommender >, < seeker >} is defined to make the model aware of the speaker's role of the next generated utterance “[Z]”, “< profile >” and “< /profile >” are special tokens that define the start and the end of the users' profile information, where “p₁, p₂, ..., p_n” highlights the users' profile and preferences, “< big5 >” and “< /big5 >” are special tokens that define the start and the end of the user Big Five personality traits, where “per₁, per₂, ..., per₅” highlights the users' Big Five personality traits values.

C. The Model Architecture

In order to utilize the users' personality traits, their preferences, and the previous conversation information during the dialogue generation process, we designed our input utterances using the *Pre-pro-prompt* template. Figure 1

highlights our PPPG-DialoGPT model architecture design. Where N defines the vocabulary size, w_j defines the j^{th} word in the vocabulary set where $j \in \{1..N\}$ and Pw_j defines the probability of generating the word j given the input data “all previous words”.

Given the prompted input x , the words within the conversation session are tokenized using the DialoGPT pre-trained tokenizer. This tokenized vector is then used to fine-tune the DialoGPT model in an autoregressive way by maximizing the probability $p(x)$ of generating the input x where at each step t , the model tries to maximize the probability $p(w_j|w_{1..w_{t-1}})$ of generating the word w_j as the next word by minimizing the CrossEntropy loss function. Then, at inference time, we use the probability logits Pw to get the most likely word as the next output token.

What sets our PPPG-DialoGPT apart from merely using prompts with other language models is the incorporation of personality traits and preferences as supplementary input information.

IV. EXPERIMENTAL SETUP

Different experiments are done to investigate the generalization performances of our models using different prompting templates. To keep the fairness of the evaluation process, all proposed models were trained using the TG-Redial dataset [6].

A. Datasets, and Training Setup

Our experiments and results use a recently published movie-based conversational recommendation dataset named *TG-Redial*. This dataset includes different information such as the users' profile, their conversation dialogue, their previous movie interactions, and the utterances' topic information to ensure a coherent dialogue and a personalized recommendation. It consists of 10000 dialogues between an information seeker and a recommender in the movie domain with specific details for the recommendation seeker such as their profile 'preferences' and the list of their previous movies interaction. We used Elourajini *et al.*'s work [35] to infer the users' Big5 personalities traits from their utterances. In their work, they employed the Electra transformer model [34] and a double-head MLP network that facilitated information sharing among different layers. This approach enabled them to predict both MBTI and Big5 personality traits concurrently from textual input data. The authors demonstrated superior performance of their model compared to various state-of-the-art models across multiple metrics, including accuracy, macro-precision, macro-recall, and macro-F1 score. Hence, we opted to adopt their solution for inferring the Big5 personality traits of users based on their utterances. We used the same training (70%), testing (15%), and validation (15%) subsets in all the experiments as well as the same hyperparameter values (50 Epochs number, Adam optimizer, a 2e-5 Learning rate, a 0.01 Weight decay, a 5 samples Batch size, and an Autoregressive loss).

For the training environment, we used the google collaborative pro version as our computing infrastructure (166.83 Gb hard

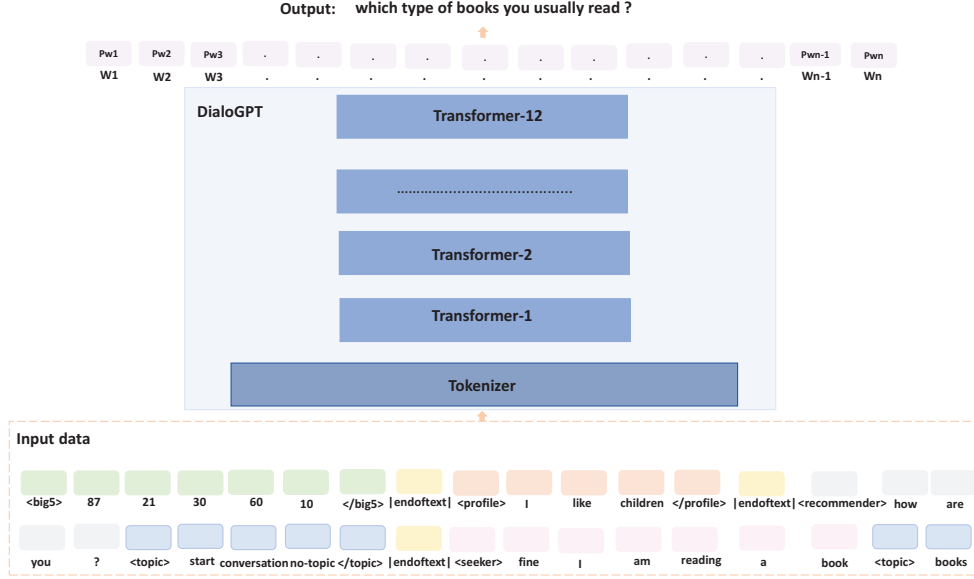


Fig. 1. PPPG-DialoGPT model architecture.

drive capacity, 25.46 GB memory capacity, and a 1 Tesla P100-PCIE GPU), which allows us to use a 20h window session of these computational resources which was sufficient to train our model for only 50 epochs. We selected the maximum number of epochs achievable (50) based on the study by Belkin *et al.* in their work titled "Reconciling modern machine learning practice and the bias-variance trade-off" [28]. In their research, they demonstrated that as the model's complexity increases (meaning more parameters and more training epochs) the likelihood of reaching the interpolating regime also increases. This results in improved model performance. Throughout our study, for every one of the 50 epochs within the training process, we compared the current validation results with the least validation loss and for each prompt-based model we kept the one that gave us the least generalization loss. Additionally, as described in Mosbach *et al.* paper [29], the best way to fine-tune a pre-trained model is to choose a small learning rate with a bias correction training algorithm to avoid vanishing gradients. Therefore we used the AdamW optimizer model with a $2e-5$ learning rate, and a 0.01 weight decay.

B. Evaluation and discussion

To assess the performance of our prompt-based models, we employed a set of commonly used automatic evaluation metrics. This included perplexity, Bert-score [30], and BLUE 1-2-3 [31] metrics, which are widely utilized for evaluating dialogue generation performance [32]. For the movie prediction performance, we used the Mean Reciprocal Rank MRR@k and the Hit Ratio HR@k metrics where $k = \{1, 3, 5, 10\}$, which are common metrics for automatically evaluating the recommendation performance. Tables II, III and IV highlight the performance of each prompt-based model compared to state-of-the-art models on

different metrics.

TABLE II
THE DIFFERENT PROMPT-BASED DIALOGPT PERFORMANCE ON THE TEST SET.

	Perplexity	BLUE-1	BLUE-2	BLUE-3	precision	recall	f1
Redial [6]	81.614	0.177	0.028	0.006	-	-	-
KBRD [6]	28.022	0.223	0.028	0.009	-	-	-
TG-Redial [6]	7.223	0.28	0.065	0.031	-	-	-
GPT-2 [6]	13.383	0.279	0.066	0.032	-	-	-
GPT-2-Topic [7]	-	0.278	0.064	0.031	-	-	-
TG-RG [7]	-	0.282	0.067	0.033	-	-	-
Stage-Two (BERT) [6]	-	0.294	0.086	0.043	-	-	-
Transformer [6]	32.856	0.287	0.071	0.032	-	-	-
Promptless	3.921	0.162	0.030	0.009	0.850	0.868	0.859
T-prompt	3.911	0.246	0.075	0.028	0.863	0.882	0.874
Pro-prompt	4.192	0.256	0.087	0.035	0.873	0.889	0.881
Pre-prompt	3.946	0.274	0.095	0.038	0.876	0.889	0.883
Per-pro-prompt	4.174	0.2668	0.0924	0.0357	0.9040	0.9161	0.9099

Table II shows that by adding more controls to the generation process both profile-based controls "Pro-prompt" and personality-based template "Pre-prompt" are outperforming the T-prompt-based model (which uses only topics as controls) almost on all evaluation metrics. This highlights the importance of using different control information to generate more convenient responses. In addition, our results show that for the BLUE@k metrics, the Pre-prompt template provides higher results compared to all of the other non-personality template-based models. This highlights the importance of using personality traits for the response generation task compared to using other types of information such as the profile information "Pro-prompt". However, we can see that combining all generation controls within one template "Pre-pro-prompt" gave us the most effective results on the BERT-score metrics. Experiments show that prompt-based leaning paradigm does affect the response generation performance on different

metrics. Using our prompt-based templates we outperformed all state-of-the-art baselines on the perplexity metric while having similar results on the BLUE-1, BLUE-2 and BLUE-3 metrics. In the Stage-Two (BERT) baseline [7] which outperforms our proposed solution on the BLUE-1 and BLUE-3 metrics, a learning architecture with 800 million trainable parameters in total is used. It combines two GPT-2-based models and one BERT-based model. However, in our proposed solution due to the prompt-based learning paradigm we only used a 117M trainable parameters model (DialoGPT-small) which is 700% less computationally expensive compared to the Stage-Two (BERT) model. Yet by using a 700% less complex model, we managed to outperform state-of-the-art models on both perplexity and BLUE-2 metrics while having almost similar results for the BLUE-1 and BLUE-3 metrics. This highlights the importance of using the prompt learning-based paradigm for enhancing both efficiency and effectiveness of the response generation model on specific metrics.

In addition to the significantly promising results on the response generation task, we were curious to know if our solution is capable of providing effective movie recommendations within the generated responses. To do that we formulated the movie prediction task as a movie generation task. When given the ground truth utterance X which contains a movie recommendation ($m \in X$ and $m \subset M$), we check if the corresponding generated response $\hat{X}$ employs the same movie as X ($m \in \hat{X}$). To ensure a fair evaluation process we choose to evaluate our models using the MRR@ k and HR@ k metrics where $k = \{1, 3, 5, 10, 50\}$. For each generation step, we encourage the model to generate k responses using the *Beam search* algorithm [33]. We consider these k generated responses as ranked predictions. Then, we define a generated response as relevant if it contains the same movie as the one mentioned in the ground truth response. Table III highlights the movie prediction “generation” results of our prompt-based models on the TG-Redial dataset compared to the existing state-of-the-art models.

TABLE III
THE DIFFERENT PROMPT-BASED DIALOGPT MOVIE PREDICTION
PERFORMANCE ON THE TEST SET.

	MRR@1	MRR@5	MRR@10	MRR@50	HR@1	HR@5	HR@10
Popularity	-	-	0.0011	0.0015	-	-	-
ReDial	-	-	0.0005	0.0009	-	-	-
KBRD	-	-	0.0040	0.0049	-	-	-
GRU4Rec	-	-	0.0014	0.0020	-	-	-
SASRec	-	-	0.0050	0.0068	-	-	-
TextCNN	-	-	0.0119	0.0133	-	-	-
BERT	-	-	0.0182	0.0221	-	-	-
TG-Redial	-	-	0.0240	0.0277	-	-	-
Promptless	0.0194	0.02681	0.0288	-	0.0194	0.0421	0.0575
T-prompt	0.0134	0.02684	0.0286	-	0.0134	0.0506	0.0644
Pro-prompt	0.0182	0.0313	0.0332	-	0.0182	0.0561	0.0712
Pre-prompt	0.0189	0.0318	0.03519	-	0.01898	0.0576	0.0850
PPPG-DialoGPT (Our's)	0.0245	0.03732	0.04080	-	0.0245	0.0635	0.0901

Experiments demonstrate that our personality-based PPPG-DialoGPT model, which is trained on an autoregressive language modeling task, outperforms state-of-the-art models across all ranking metrics. This highlights the effectiveness of prompt-based learning for transferring the knowledge of

text generation models to other tasks, such as the movie recommendation task. Also, table III illustrates that the personality-based models “Pre-prompt, and Pro-pre-prompt models” are outperforming all of the other prompt-based baselines on the movie recommendation task. This highlights the importance of including the individuals’ personality traits for the session-based recommendation problems. Moreover, another outstanding result that we obtained during our experiments is that by only producing 10 ranked results (MRR@10), our PPPG-DialoGPT model outperforms state-of-the-art models on the MRR@50 metric. This significantly improves the efficiency of the TG-Redial movie prediction task compared to existing baselines as we have managed to predict more correct movies by considering less ranked predicted samples. Despite the promising results provided by employing all of the individual Big Five personality traits on the movie recommendation task, we were curious to investigate the impact of each trait on the ranking results independently. In order to do this, we managed to select a specific personality trait within the personality control tokens “< big5 > ..., < /big5 >” and we changed its original value with the DialoGPT tokenizer unknown token while keeping all the other traits unchanged. By doing this we succeeded in investigating the impact of each personality trait’s absence on the movie recommendation task. Table IV highlights the impact of each personality trait absence on the MRR@ k and HR@ k metrics.

Although there is a very slight difference between adding

TABLE IV
THE IMPACT OF PERSONALITY TRAITS ON THE MOVIE RECOMMENDATION
TASK.

	MRR@1	MRR@5	MRR@10	HR@1	HR@5	HR@10
PPPG-DialoGPT	0.0245	0.03732	0.04080	0.0245	0.0635	0.0901
Removed AGR	0.0247	0.0377	0.0417	0.0247	0.0628	0.0943
Difference: with AGR	+0.0002	+0.00038	+0.0009	+0.0002	-0.0007	+0.0042
Removed CSN	0.0251	0.0370	0.0416	0.0251	0.0585	0.0929
Difference: with CSN	+0.0006	-0.0003	+0.0008	+0.0006	-0.005	+0.0028
Removed EXT	0.0255	0.0381	0.0417	0.0255	0.0632	0.0913
Difference: with EXT	+0.001	+0.00078	+0.0009	+0.001	-0.0003	+0.0012
Removed NEU	0.0246	0.0380	0.0416	0.0246	0.0637	0.0917
Difference: with NEU	+0.0001	+0.00068	+0.0008	+0.0001	+0.0002	+0.0016
Removed OPN	0.0238	0.0373	0.0414	0.0238	0.0619	0.0929
Difference: with OPN	-0.0007	-0.00002	+0.0006	-0.0007	-0.0016	+0.0028

and removing a specific trait from the user’s personality, experiments show that removing the Big Five *Openness* (OPN [14]) trait from the structure-based input led to a small decrease in the performance of our PPPG-DialoGPT model on almost all of the ranking metrics. The reason behind this is that OPN traits reflect a person’s level of imagination, creativity, and insightfulness. Consequently, having a high degree of openness suggests that the individual is receptive to new movie recommendations. As a result, the set of possible movies that can be suggested to such a user is larger compared to individuals with lower levels of openness which increases the likelihood of recommending the right movie to them. This highlights the importance of having the individual’s OPN trait to effectively provide convenient and personalized movie recommendations compared to the other

personality traits.

V. CONCLUSION

In this paper, we propose a novel framework for recommendations based on personality traits and the use of a prompt-based learning method. The experimental results showed that our PPPG-DialoGPT model yields promising results on both response generation and movie recommendation tasks compared to state-of-the-art models. Moreover, our experiments show that Big Five "OPN" trait have the highest impact on effectively producing the right recommendation compared to the other traits.

In future work, we aim to investigate the impact of using automatic prompt-learning techniques instead of manually designed prompts. Also, we are curious to investigate removing more than one trait at a time. Additionally, since dialogue can be very long and DialoGPT input text is limited to 1024 sub-words, our next objective is to explore the use of larger generative models such as ChatGPT. Our proposed method has great potential for personalized recommendation systems and can be further extended to other domains.

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